
3AL Tutorial 10 2024 Solutions

1. Let G be a group with order $p^n m$ where $n \geq 1$, p is a prime with $p \nmid m$. Suppose H is a Sylow p -subgroup of G .
- (a) Prove that H is a Sylow p -subgroup of $N_G(H)$ and that it is the only Sylow p -subgroup of $N_G(H)$.

Suppose $|G| = p^n m$, where p does not divide m . Then $|H| = p^n$. We know that H is a subgroup of $N_G(H)$. Thus $p^n \mid |N_G(H)|$. We know that $|N_G(H)| \mid p^n m$. Thus $N_G(H) = p^n k$ for some k where p does not divide k . Thus H is a Sylow p -subgroup of $N_G(H)$. Moreover, since H is normal in $N_G(H)$, it is the unique Sylow p -subgroup of $N_G(H)$.

- (b) Let $a \in G$ such that $|a| = p^k$ where $k \geq 1$. Prove that if $aHa^{-1} = H$, then $a \in H$.

If $aHa^{-1} = H$, then $a \in N_G(H)$. Then $\langle a \rangle$ is a p -subgroup of $N_G(H)$ and so it is contained in a Sylow p -subgroup of $N_G(H)$. However, H is the only Sylow p -subgroup and a Sylow p -subgroup is a maximal p -subgroup. Therefore, $\langle a \rangle$ must be contained in H and so $a \in H$.

2. Let G be a finite group and H a normal p -subgroup. Show that if K is a Sylow p -subgroup of G , then H is a subgroup of K .

Since H is a p -subgroup, there is some Sylow p -subgroup of G , say K' , such that $H \subseteq K'$. By the Second Sylow Theorem, since K and K' are Sylow p -subgroups, there is some $g \in G$ such that $gK'g^{-1} = K$. Since $H \subseteq K'$, we have $gHg^{-1} \subseteq K$. However, H is normal so we have $H = gHg^{-1} \subseteq K$. Thus H is a subgroup of K .

3. Prove the following:

Frattini's Argument. *Let G be a finite group with normal subgroup H . If K is a Sylow p -subgroup of H , then $G = HN_G(K)$.*

HINT: Since H is normal in G , we have $gKg^{-1} \subseteq H$ for all $g \in G$. Explain why this is true.

For simplicity, we write N for $N_G(K)$. We already know that HN is a subset of G so it remains to prove that $G \subseteq HN$. Take $g \in G$. Since H is normal in G and K is a subgroup of H , we have $gKg^{-1} \subseteq H$. Since gKg^{-1} is a group of the same order as K , we have that gKg^{-1} is another Sylow p -subgroup of H . By the Second Sylow Theorem, there is some $h \in H$ such that $h(gKg^{-1})h^{-1} = K$. This means that $hg \in N_G(K) = N$. Therefore, we can write $g = h^{-1}(hg) \in HN$. Thus $G = HN_G(K)$, as desired.

4. Let G be a finite group and p be a prime that divides $|G|$. Prove or disprove: If H and K are Sylow p -subgroups then H and K are isomorphic.

If H and K are Sylow p -subgroups, then by the Second Sylow Theorem, there is some $g \in G$ such that $K = gHg^{-1}$. Consider the map $\varphi : H \rightarrow K$ defined by $\varphi(h) = ghg^{-1}$. Since $K = gHg^{-1}$, the map is surjective. Moreover, if $\varphi(h) = \varphi(h')$ then $ghg^{-1} = gh'g^{-1}$ and so $h = h'$. Thus φ is injective. Finally, for $h, h' \in H$, we have $\varphi(hh') = gh'hg^{-1} = ghg^{-1}gh'g^{-1} = \varphi(h)\varphi(h')$. Therefore, φ is an isomorphism.

5. Let G be a group of order $p^n m$ where $n \geq 1$ and p is a prime that does not divide m . Suppose that H is a normal subgroup of G such that $|G : H|$ is coprime to p . Show that H contains all Sylow p -subgroups of G .

Since $|G : H|$ and p are coprime, we must have $|G/H| = d$ for some d that divides m . That is, $|H| = p^n(m/d)$. Let K be a Sylow p -subgroup of G , we will show that $K \subseteq H$. By the First Sylow Theorem, there is some Sylow p -subgroup P of H . It must have order p^n and so it is also a Sylow p -subgroup of G . By the Second Sylow Theorem, there is some $g \in G$ such that $K = gPg^{-1}$. However, $P \subseteq H$ and H is normal in G . So $K = gPg^{-1} \subseteq gHg^{-1} = H$. Thus every Sylow p -subgroup K of G is contained in H .

6. Let G be a group of order $90 = 2 \times 3^2 \times 5$. Suppose P is a Sylow 5-subgroup of G that is not normal. Determine the number of conjugates of P .

The number of conjugates of Sylow 5-subgroup is the number of Sylow 5-subgroups. That is, n_5 . By the Third Sylow Theorem, $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 18$. Thus either $n_5 = 1$ or $n_5 = 6$. Since P is not normal, it is not a unique Sylow 5-subgroup and so $n_5 \neq 1$. Thus there are 6 Sylow 5-subgroups and 6 conjugates of P (including itself).

7. Prove that a group of order 200 has a nontrivial proper normal subgroup.

Let G be a group of order $200 = 2^3 \times 5^2$. By the Third Sylow Theorem, $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 8$. So $n_5 = 1$ and there is a unique Sylow 5-subgroup. Furthermore, this subgroup is normal.

8. Use the Third Sylow Theorem to show that a group of order 18 has a nontrivial proper normal subgroup.

Let G be a group of order $18 = 2 \times 9$. By the Third Sylow Theorem, $n_9 \equiv 1 \pmod{3}$ and $n_9 \mid 2$. Thus $n_9 = 1$ and so there is a unique Sylow 3-subgroup of order 9. Since

$9 \neq 1, 18$, this is a nontrivial proper normal subgroup.

9. Let G be a group of order $56 = 2^3 \times 7$.

- (a) Determine the possibilities for n_2 and n_7 .
- (b) Determine the number of elements of order 7 if $n_7 \neq 1$.
- (c) Show that either $n_2 = 1$ or $n_7 = 1$.

(a) By the Third Sylow Theorem, $n_2 \equiv 1 \pmod{2}$ and $n_2 \mid 7$. Thus either $n_2 = 1$ or $n_2 = 7$. Similarly, $n_7 \equiv 1 \pmod{7}$ and $n_7 \mid 8$. Thus $n_7 = 1$ or $n_7 = 8$.

(b) Suppose $n_7 \neq 1$. This means that $n_7 = 8$. Then there are $8 \times (7 - 1) = 8 \times 6 = 48$ elements of order 7. [If this was a test or exam, ideally include reasoning why no Sylow 7-subgroups have any elements of order 7 in common]

(c) If $n_7 = 1$, we are done. On the other hand, if $n_7 = 8$, there are $56 - 48 = 8$ elements of order other than 7. A Sylow 2-subgroup consists of 8 elements. So, there can only be one Sylow 2-subgroup. That is, $n_2 = 1$. (Note that Sylow 2-subgroups cannot contain any element of order 7 by Lagrange's Theorem).

Extra problems

- (I) Let P be a Sylow p -subgroup of a group G and H be a subgroup of G such that $N_G(P) \subseteq H$. Show that $N_G(H) = H$.

HINT: Show that, for $a \in N_G(H)$, we have $aPa^{-1} \subseteq H$. Then use the Second Sylow Theorem.

It is clear that $H \subseteq N_G(H)$, so it remains to show $N_G(H) \subseteq H$. Take $a \in N_G(H)$, then $aHa^{-1} = H$. By assumption, we have $P \subseteq N_G(P) \subseteq H$. Thus $aPa^{-1} \subseteq aHa^{-1} = H$. So aPa^{-1} is a conjugate of P and it is another Sylow p -subgroup of G . However, it must also be a Sylow p -subgroup of H . Similarly, P is a Sylow

p -subgroup of H . So, by the Second Sylow Theorem, there is some $h \in H$ such that $h(aPa^{-1})h^{-1} = P$. That is, $ha \in N_G(P) \subseteq H$. Since $h \in H$ and H is a group, we must have $h^{-1}(ha) = a \in H$. Thus $N_G(H) = H$.